IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Iryna Rakhuba
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Data Collection
- Data Wrangling
- Exploratory Data Analysis with SQL
- Exploratory Data Analysis with Data Visualization
- Interactive Visual Analytics
- Building an Interactive Map with Folium
- Building an Interactive Dashboard with Plotly Dash
- Predictive Analysis (Classification)

Summary of results

- Exploratory Data Analysis Results
- Interactive Dashboard Screenshots
- Results of the Predictive Analysis

Introduction

It’s a project to predict whether the Falcon 9 rocket's first stage will land successfully. Why does this matter? SpaceX can reuse these rocket stages, which helps them save over 100 million dollars on each launch — making them much cheaper than other companies. Our goal is to create a model that can predict a successful landing. This kind of analysis shows how data can help make smart business decisions in the aerospace industry.

To achieve the objective, we'll undertake the following key tasks:

- Identify relevant launch data features
- Engineer new features for prediction
- Transform data for model suitability
- Develop predictive models for landing success

Section 1

Methodology

Methodology

Data collection methodology:

- Request to the SpaceX REST API.
- WEb scraping from Wikipedia’s page “List of Falcon 9 and Falcon heavy launches”

Perform data wrangling

- Data filtering
- Handled missing with imputation
- Converting categorical variables (one-hot encoding)

Perform exploratory data analysis (EDA) using visualization and SQL

Perform interactive visual analytics using Folium and Plotly Dash

Perform predictive analysis using classification models

(SVM, Classification tree, Logistic Regression)

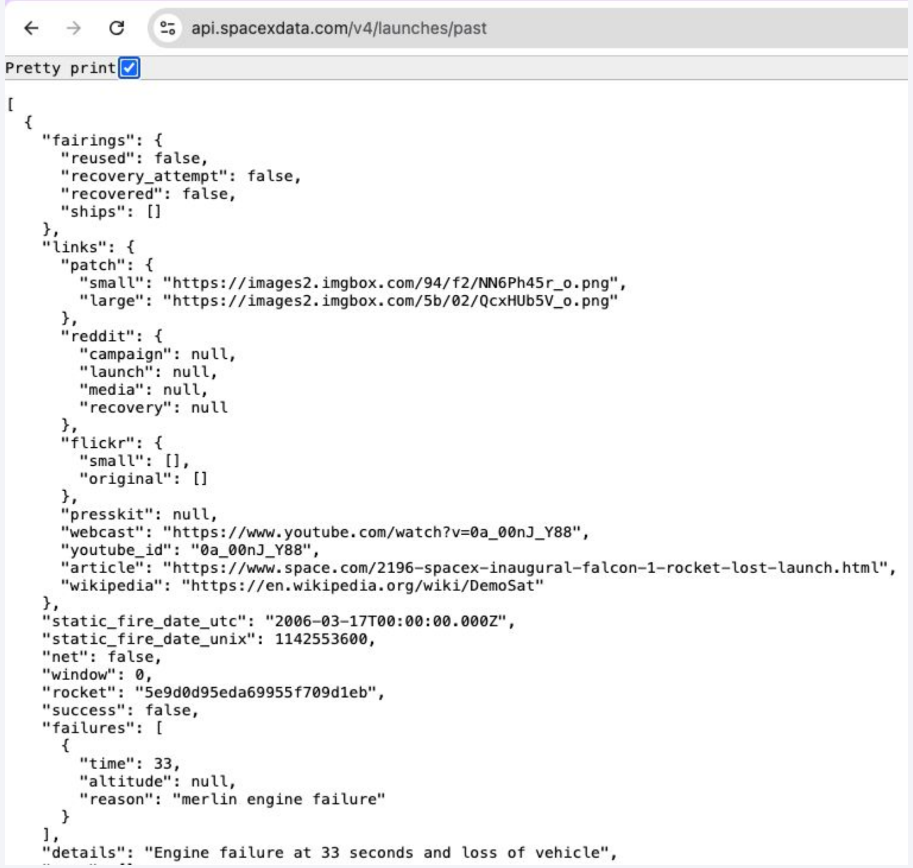
- Construction, tuning, validation of models

Data Collection

Data sets were collected from two sources:

1 - SpaceX API

2 - Wiki Pages

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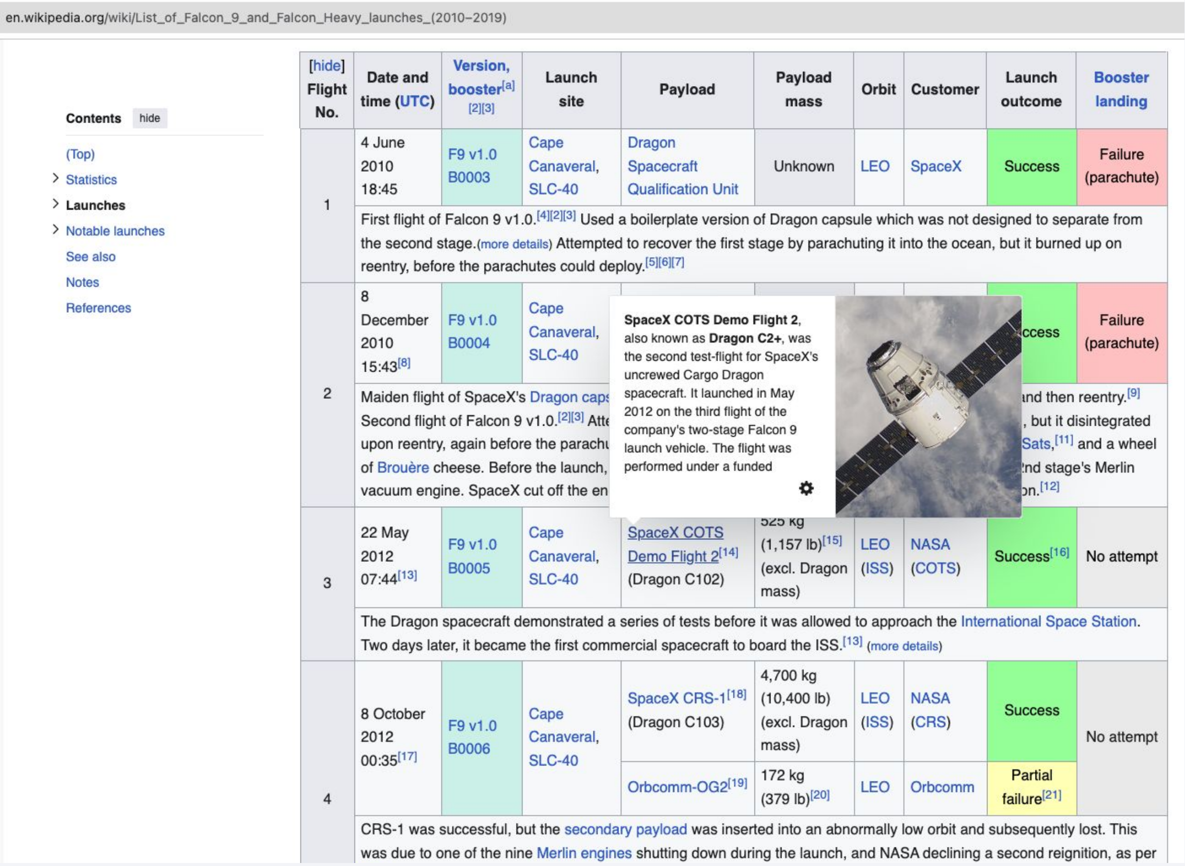


[hide] Flight No.	Date and time (UTC)	Version, booster[a] [2][3]	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Booster landing
1	4 June 2010 18:45	F9 v1.0 B0003	Cape Canaveral, SLC-40	Dragon Spacecraft Qualification Unit	Unknown	LEO	SpaceX	Success	Failure (parachute)
	First flight of Falcon 9 v1.0.[4][2][3] Used a boilerplate version of Dragon capsule which was not designed to separate from the second stage.(more details) Attempted to recover the first stage by parachuting it into the ocean, but it burned up on reentry, before the parachutes could deploy.[5][6][7]								
2	8 December 2010 15:43[8]	F9 v1.0 B0004	Cape Canaveral, SLC-40					ccess	Failure (parachute)
	Maiden flight of SpaceX's Dragon caps … nd then reentry.[9] Second flight of Falcon 9 v1.0.[2][3] Atte … , but it disintegrated upon reentry, again before the parachu … Sats,[11] and a wheel of Brouère cheese. Before the launch, … nd stage's Merlin vacuum engine. SpaceX cut off the en … on.[12]								
3	22 May 2012 07:44[13]	F9 v1.0 B0005	Cape Canaveral, SLC-40	SpaceX COTS Demo Flight 2[14] (Dragon C102)	525 kg (1,157 lb)[15] (excl. Dragon mass)	LEO (ISS)	NASA (COTS)	Success[16]	No attempt
	The Dragon spacecraft demonstrated a series of tests before it was allowed to approach the International Space Station. Two days later, it became the first commercial spacecraft to board the ISS.[13] (more details)								
4	8 October 2012 00:35[17]	F9 v1.0 B0006	Cape Canaveral, SLC-40	SpaceX CRS-1[18] (Dragon C103)	4,700 kg (10,400 lb) (excl. Dragon mass)	LEO (ISS)	NASA (CRS)	Success	No attempt
				Orbcomm-OG2[19]	172 kg (379 lb)[20]	LEO	Orbcomm	Partial failure[21]	
	CRS-1 was successful, but the secondary payload was inserted into an abnormally low orbit and subsequently lost. This was due to one of the nine Merlin engines shutting down during the launch, and NASA declining a second reignition, as per								

Data Collection – SpaceX API

The SpaceX REST API provides next data on launches:

- rocket details
- payloads
- landing outcomes.

Send API Request to SpaceX API
↓
Parse JSON Response
↓
Extract Relevant Data to DataFrame
↓
Filter the DataFrame
↓
Transform and Clean the Data
↓
Export the Data into a CSV File

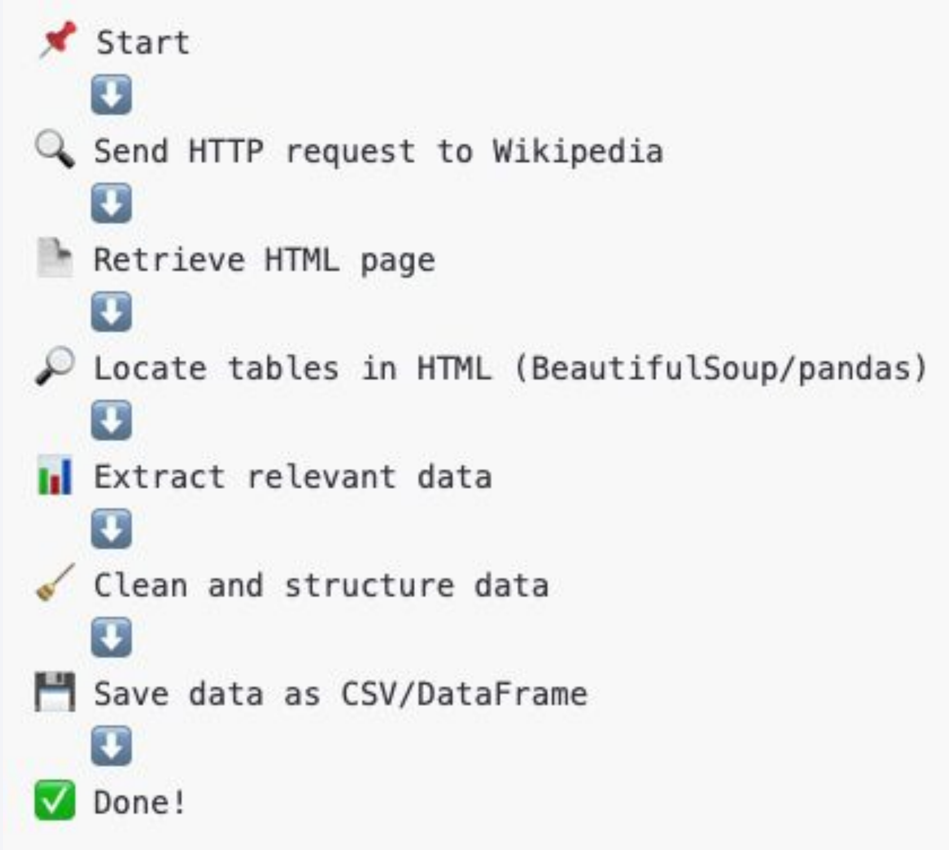
api.spacexdata.com/v4/launches/past

```
spacex_url="https://api.spacexdata.com/v4/launches/past"
response = requests.get(spacex_url)
response.status_code
data_j = response.json()
data = pd.json_normalize(data_j)
data = data[['rocket', 'payloads', 'launchpad', 'cores',
'flight_number', 'date_utc']]
data = data[data['cores'].map(len)==1]
data = data[data['payloads'].map(len)==1]
data['cores'] = data['cores'].map(lambda x : x[0])
data['payloads'] = data['payloads'].map(lambda x : x[0])
data['date'] = pd.to_datetime(data['date_utc']).dt.date
data = data[data['date'] <= datetime.date(2020, 11, 13)]
```

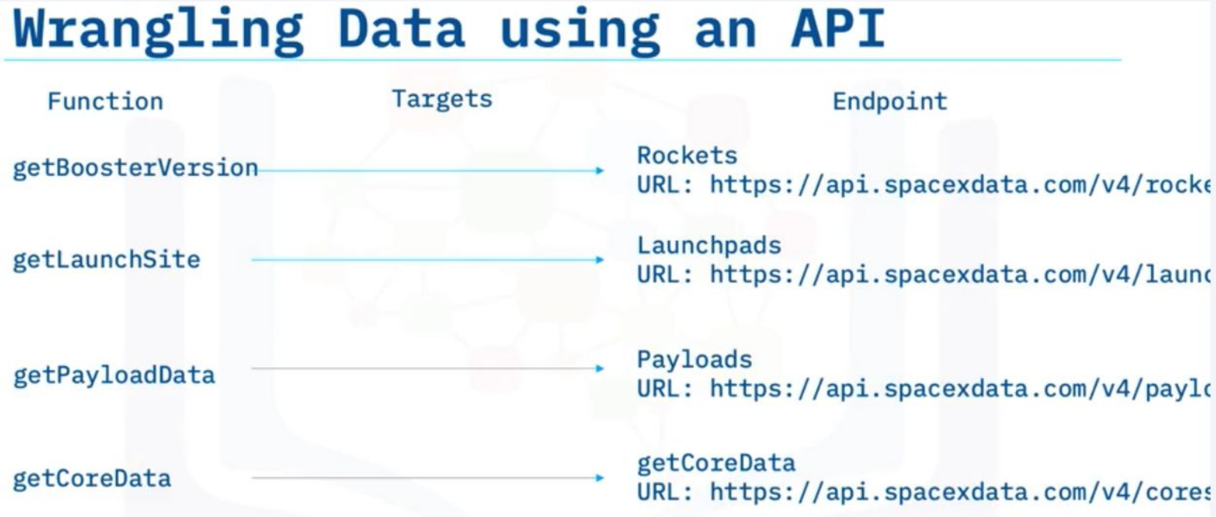
Data Collection - Scraping

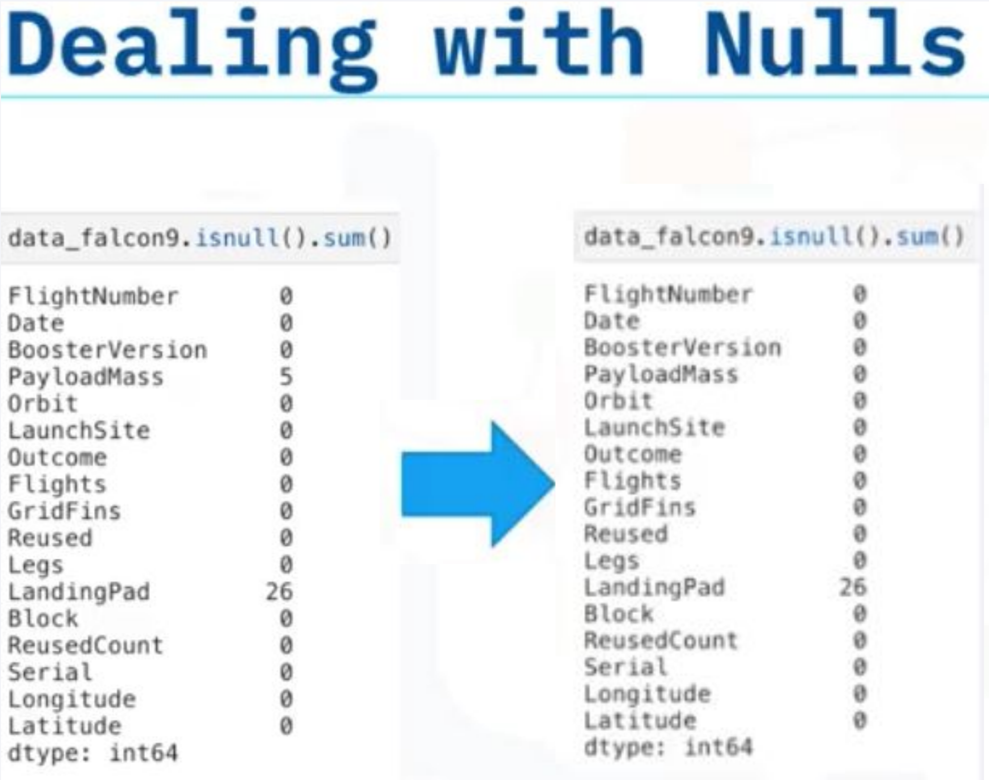
Key Steps of Web Scraping

1. **Send Request** → Use `requests` to get the HTML page.
2. **Parse HTML** → Use `BeautifulSoup` to extract relevant data.
3. **Extract Tables** → Use `pandas.read_html()` to retrieve tables.
4. **Clean and Transform Data** → Convert it into a structured format.
5. **Save Data** → Store it in CSV or a DataFrame.



Data Wrangling





- Identify and calculate the percentage of the missing values in each attribute
- Identify which columns are numerical and categorical
- Deal with Nulls

EDA with Data Visualization

Following charts were generated to visually represent key aspects of our data

- Flight Number vs. Payload Mass
- Flight Number vs. Launch Site
- Flight Number vs. Orbit Type
- Payload Mass vs. Orbit Type
- Success Rate of Each Orbit Type
- Yearly Trend of Success Rate

Scatter plots excel at showing the relationship between two numerical variables. They can reveal correlations, patterns, and trends that might not be apparent in other chart types.

Bar charts are ideal for comparing values across different categories. They make it easy to see which category has the highest or lowest value

Line charts are excellent for showing trends and changes over time. They can illustrate how a variable changes over a continuous period

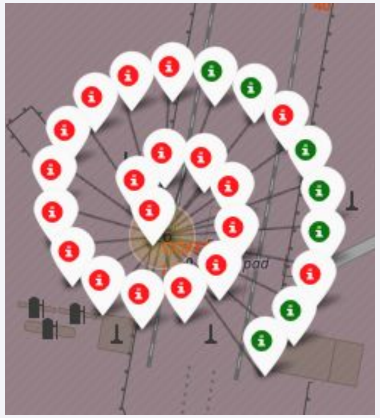
EDA with SQL

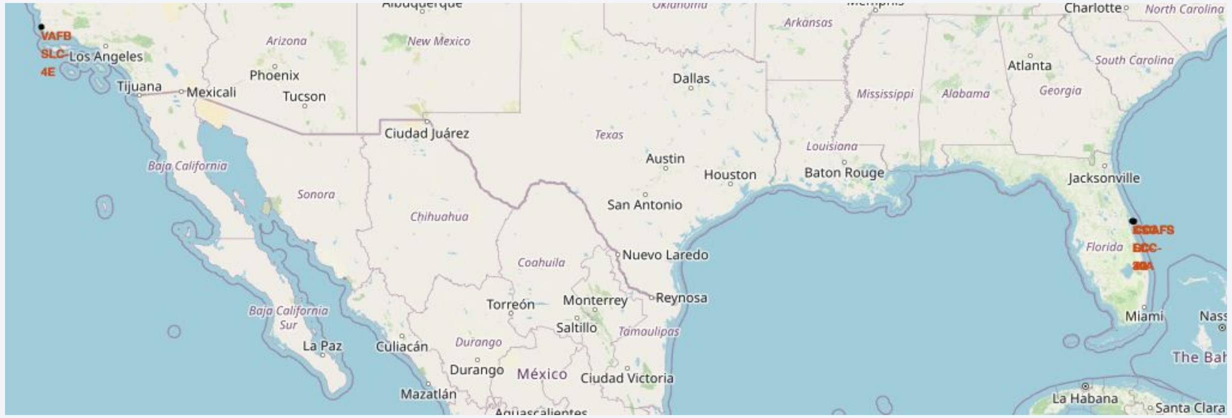
- %sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE
- %sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE '%CCA%' LIMIT 5
- %sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
- %sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version LIKE '%F9 v1.1%'
- %sql select MIN(Date) from SPACEXTABLE where Landing_Outcome = 'Success (ground pad)'
- %sql select Booster_Version from SPACEXTABLE where Landing_Outcome = 'Success (drone ship)' and PAYLOAD_MASS__KG_ between 4000 and 6000
- %sql select COUNT(Mission_Outcome) from SPACEXTABLE
- %sql select Booster_Version, PAYLOAD_MASS__KG_ from SPACEXTABLE where PAYLOAD_MASS__KG_=(select MAX(PAYLOAD_MASS__KG_)from SPACEXTABLE)
- %sql SELECT substr(Date, 6,2) as Month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE where Landing_Outcome='Failure (drone ship)' and substr(Date,0,5)='2015'
- %sql select Landing_Outcome, Date, COUNT(*) as Landing_Outcome_count from SPACEXTABLE where Date between '2010-06-04' and '2017-03-20' GROUP BY Landing_Outcome ORDER BY Landing_Outcome_count DESC

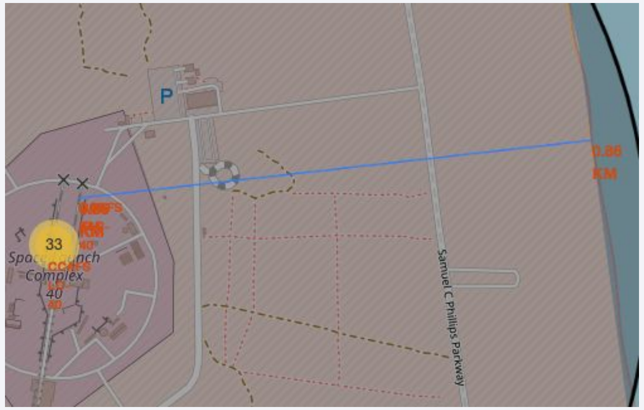
EDA with SQL on GitHub by Iryna Rakhuba

Build an Interactive Map with Folium

- All launch sites were marked on a map → folium.Circle(), folium.Marker()
- The success/failed launches for each site were marked on the map → folium.Marker(), MarkerCluster()
- The distances between a launch site to its proximities were calculated → MousePosition(), folium.PolyLine()







Build a Dashboard with Plotly Dash

Dashboard allows analyze launch records interactively.

- **Dropdown List for Launch Sites** to select and filter data by individual launch sites or view data for all sites combined
- **Pie Charts for Launch Sites** to visualize the distribution of launch outcomes for both individual and all launch sites
- **Scatter Plot with Payload Mass Slider** to explore the relationship between payload mass and launch outcomes

Predictive Analysis (Classification)

1. Building / Improving

Determine Training Labels → *to_numpy()*

Standardize the Data → *preprocessing.StandardScaler()*

Split into Training Data and Test Data → *train_test_split()*

Find Best Hyperparameter GridSearchCV() for:

SVM → *SVC()*

Classification Trees → *DecisionTreeClassifier()*

Logistic Regression → *LogisticRegression()*

↓

2. Evaluating

calculate the accuracy → *score(), classification_report()*

↓

3. Selecting of the best model

comparison of confusions matrixes → *plot_confusion_matrix()*

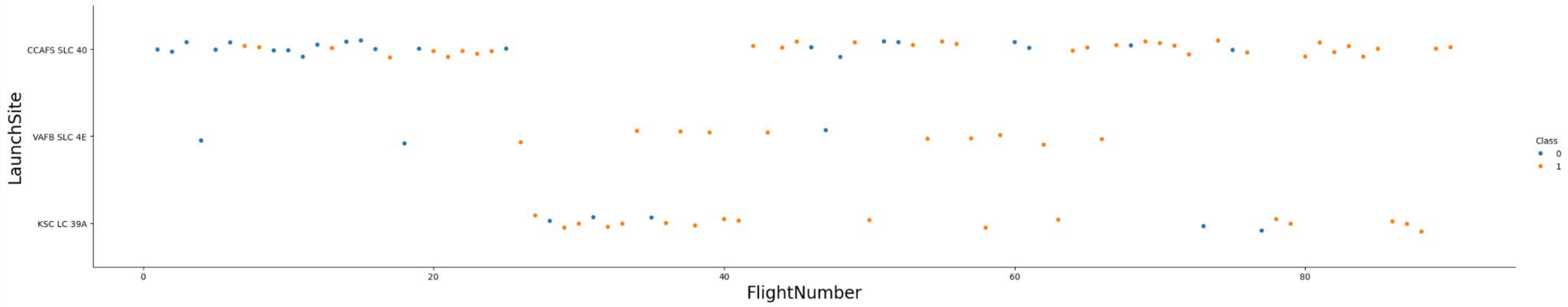
Results

- **Exploratory data analysis** shows that success rate can be affected by next features:
  - FlightNumber
  - PayloadMass
  - Orbit
  - LaunchSite
  - Year
- **Interactive analytics demo in screenshots**
  - KSC LC-39A has the highest success rate (41.7%), making it the most reliable launch site
  - CCAFS LC-40 follows with 29.2% of successful launches
  - VAFB SLC-4E and CCAFS SLC-40 have lower success percentages (16.7% and 12.5%)
  - The dashboard allows filtering by launch site, helping analyze performance trends individually
- **Predictive analysis results**
  - All 4 model demonstrate the same confusion matrix
  - Each of them can distinguish between the different classes

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

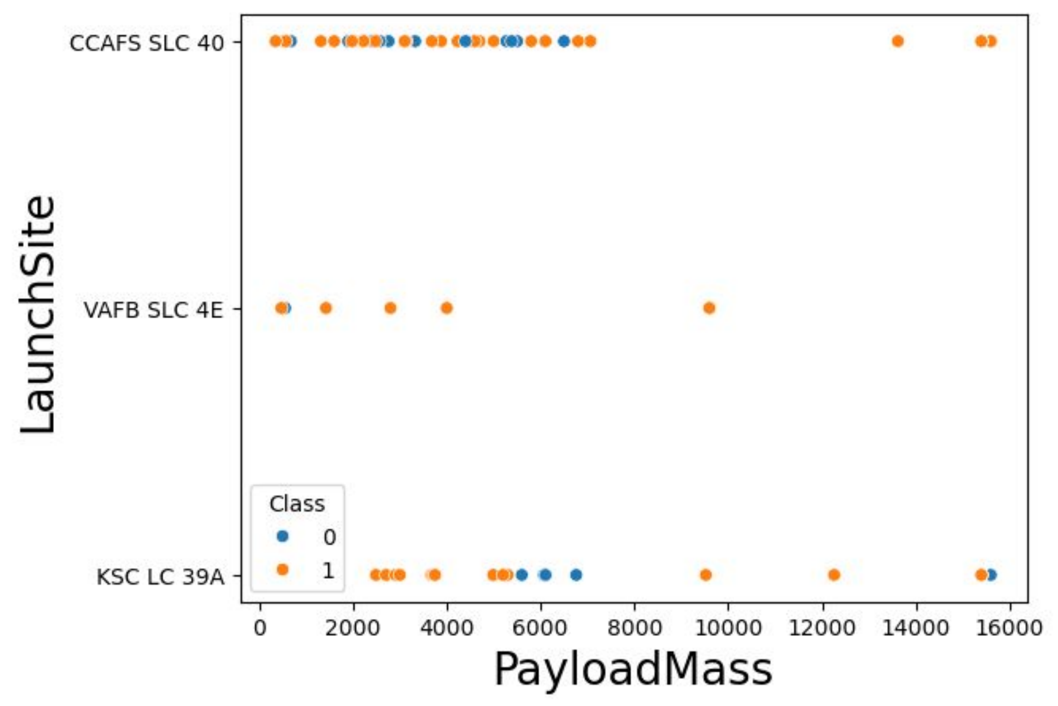


1. **Success vs. Failure Distribution**
    - The plot shows the relationship between **Flight Number** and **Launch Site**, with color representing mission success (1 - orange) and failure (0 - blue)
    - Over time (higher Flight Number), **the success rate appears to improve**, as there are more orange dots towards the right
2. **Launch Site Comparison**
    - **CCAFS SLC 40** has the highest number of launches, with a mix of successes and failures.
    - **VAFB SLC 4E** has fewer launches, with failures in early launches but more successes later.
    - **KSC LC 39A** has a noticeable trend of increasing success after initial failures.
3. **General Trend**
    - **Early flights had more failures**, suggesting an improvement in reliability over time.
    - **Each launch site shows a learning curve**, with failures initially and more successful launches as experience increases.

This suggests that SpaceX improved its launch technology and operations over time, increasing the likelihood of success as they gained experience with more flights.

Payload Mass vs. Launch Site

1. **Success vs. Failure by Payload Mass**
    - The plot shows the relationship between **Payload Mass** and **Launch Site**, with color representing mission success (1 - orange) and failure (0 - blue).
    - **Higher payloads tend to have more successful launches**, especially above 10,000 kg.
2. **Launch Site Comparison**
    - **CCAFS SLC 40** has a wide range of payloads, with both successes and failures, but **failures seem more common for medium-range payloads (4000-6000 kg)**.
    - **VAFB SLC 4E** shows fewer launches overall, but most seem successful, especially for lower payloads (<6000 kg).
    - **KSC LC 39A** has **fewer failures for payloads up to 4000 kg**, but failures appear at higher payloads (~6000 kg and above)

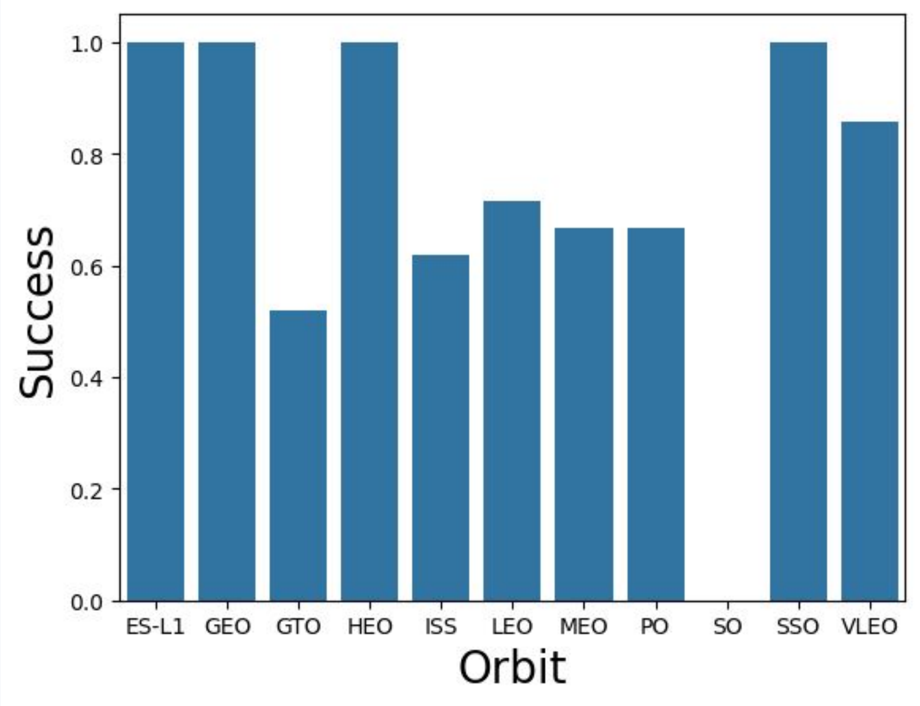


3. **General Trends**
    - **Lighter payloads (0-4000 kg)**: High success rates across all launch sites.
    - **Medium payloads (4000-6000 kg)**: More failures, particularly at CCAFS SLC 40 and KSC LC 39A.
    - **Heavy payloads (above 10,000 kg)**: Almost all successful, indicating that SpaceX has optimized launches for very high payloads.

This suggests that mission success is influenced by payload weight, with failures occurring more frequently in the medium payload range but improving significantly for very heavy payloads.

Success Rate vs. Orbit Type

1. **High Success Rates for Specific Orbits**
   - **ES-L1, GEO, HEO, and SSO have a 100% success rate**.
   - This suggests that **missions targeting these orbits are well-optimized and reliable**.
2. **Lower Success for GTO (Geostationary Transfer Orbit)**
   - GTO shows the **lowest success rate**, around **50-60%**.
   - This could be due to the **complexity and energy requirements of transferring payloads from GTO to GEO**.
   - Failures may also be related to **booster performance or payload deployment issues**.
3. **Moderate Success Rates for ISS, LEO, MEO, and PO**
   - ISS, LEO (Low Earth Orbit), MEO (Medium Earth Orbit), and PO (Polar Orbit) have **success rates between 60-80%**.
   - LEO and ISS missions likely include **cargo resupply and satellite deployments**, which have seen **some failures but mostly stable performance**.
   - MEO and PO might involve **navigation and remote sensing satellites**, which have **higher risks compared to purely low-Earth orbit launches**.
4. **VLEO (Very Low Earth Orbit) and SO (Sun-Synchronous Orbit) Perform Well**
   - **VLEO has a success rate above 80%**, which is a **good indicator for small satellite and reconnaissance missions**.
   - **SSO (Sun-Synchronous Orbit) reaches 100%**, showing strong reliability for **Earth observation and scientific missions**.



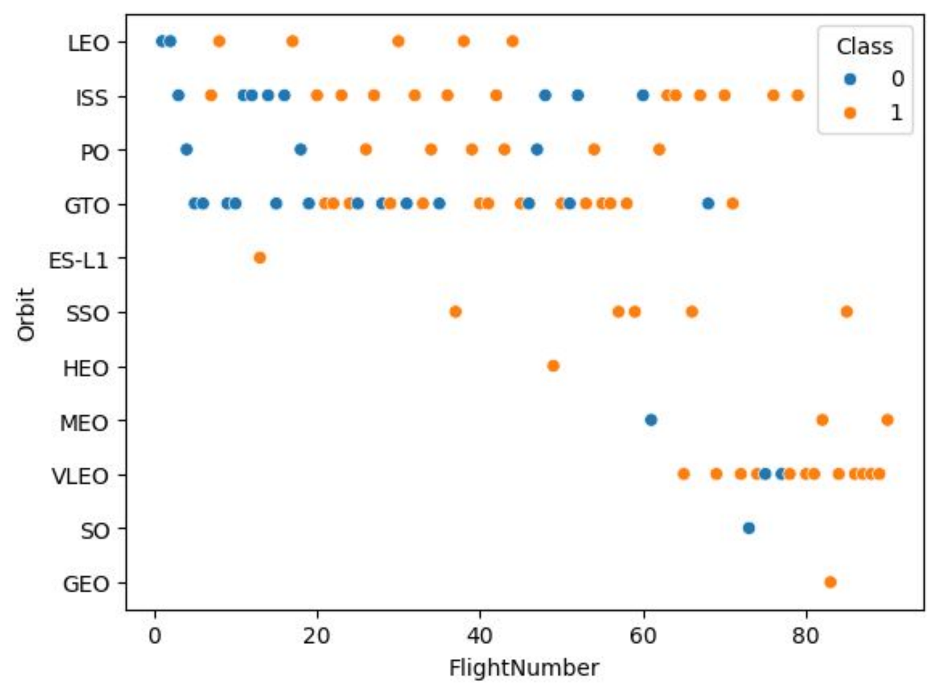
Insights & Implications:

- SpaceX has **mastered missions to GEO, HEO, SSO, and ES-L1**.
- **GTO is the riskiest orbit** in this dataset.
- **ISS and LEO have moderate risks**, likely due to payload complexities.
- The overall trend suggests **higher-altitude orbits (SSO, GEO, HEO) tend to have better success rates compared to lower-altitude orbits like LEO, MEO, and GTO**.

Conclusion: SpaceX's launch success varies by orbit, with GTO being the most challenging, while GEO, SSO, and deep space missions show strong reliability.

Flight Number vs. Orbit Type

1. **Higher Flight Numbers Show More Success (Class 1 - Orange)**
   - As **Flight Number** increases, we see a higher concentration of **orange dots**, indicating that success rates improve over time.
   - This suggests that SpaceX has improved its landing technology and execution with experience.
2. **Orbit Type Affects Success Rate**
   - **LEO (Low Earth Orbit), ISS (International Space Station), and PO (Polar Orbit)** have a mix of both successes (orange) and failures (blue).
   - **GTO (Geostationary Transfer Orbit)** has more failures, possibly due to the increased difficulty of reaching this orbit.
   - **VLEO (Very Low Earth Orbit) and SO (Solar Orbit)** show a higher percentage of successful landings.
3. **More Frequent Launches to Certain Orbits**
   - The densest areas of the scatter plot are in **LEO, ISS, PO, and GTO**, indicating that most missions target these orbits.
   - Other orbits (ES-L1, SO, GEO) have fewer data points, meaning SpaceX attempts them less frequently.

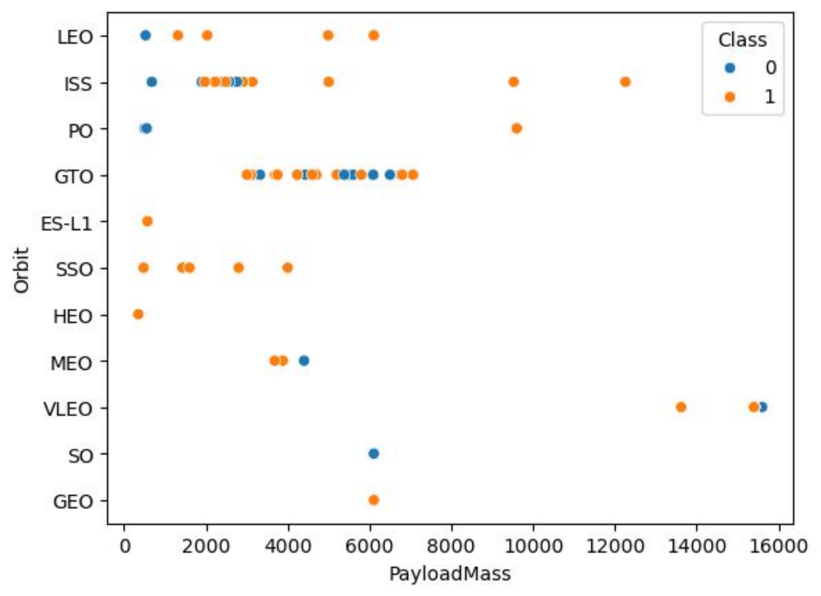


Conclusion:

- SpaceX has improved its landing success rate over time.
- The success rate varies by orbit, with higher success in **LEO, ISS, and VLEO** and more failures in **GTO and HEO**.
- Frequent mission types (LEO, ISS) benefit from iterative learning, leading to better landing performance.

Payload vs. Orbit Type

1. **Higher Payload Mass Still Achieves Success**
   - There are successful landings (orange) even for very high payloads (~15,000 kg).
   - This suggests that SpaceX has improved its ability to land boosters successfully despite carrying heavier payloads.
2. **Orbit Type Influences Success Rate**
   - **LEO, ISS, and PO** show a high success rate across a range of payload masses.
   - **GTO (Geostationary Transfer Orbit)** has a mix of successes and failures, particularly between 4,000–6,000 kg.
   - **HEO (Highly Elliptical Orbit) and SSO (Sun-Synchronous Orbit)** have fewer data points, but the pattern suggests moderate success.
3. **Lower Payloads Have Mixed Results**
   - At lower payloads (~0–6,000 kg), there is a mix of both successes (orange) and failures (blue), indicating that mission success at lower payloads was not always guaranteed in earlier flights.

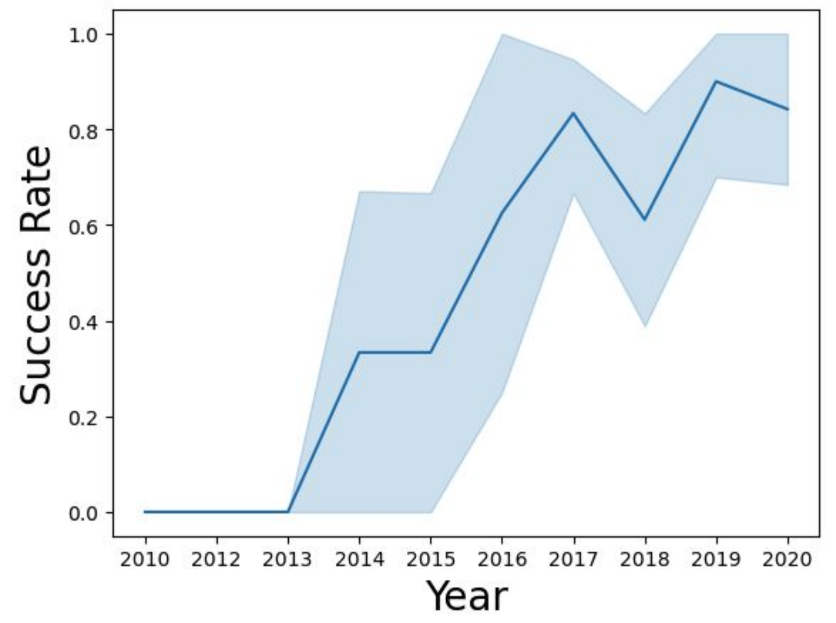


Conclusion:

- **SpaceX has improved success rates even for heavy payloads.**
- **LEO and ISS missions are more reliable**, while **GTO and HEO show more failures**.
- **Earlier flights and lighter payloads had more failures**, but experience has improved outcomes.

Launch Success Yearly Trend

1. **Steady Increase in Success Rate Over Time**
   - In the early years (2010–2014), most missions resulted in failures (Class close to 0).
   - As time progresses, the success rate improves, with most missions succeeding after 2016.
   - By 2019–2020, the majority of missions are successful (Class close to 1).
2. **Regression Line Shows a Strong Positive Trend**
   - The best-fit line indicates a **consistent improvement** in success rates.
   - The confidence interval (shaded region) is wider at the beginning (uncertainty in early missions) and narrows as success becomes more predictable.



3. **SpaceX's Learning Curve & Technological Improvements**
   - The increasing success rate aligns with SpaceX's advancements in **Falcon 9 reusability, landing technology, and operational experience**.
   - The failures in early years suggest experimental phases, while later years reflect **refined landing techniques**.

Conclusion:

- **SpaceX's success rate has significantly improved over time**.
- **Failures were common in early missions, but after 2015, success became the norm**.
- **The trend suggests that SpaceX's iterative learning approach and technological advancements have led to higher mission reliability**.

All Launch Site Names

SELECT DISTINCT Launch_Site
FROM SPACEXTABLE

The `DISTINCT` keyword in SQL is used to retrieve unique (distinct) values from a column in a table. It helps remove duplicate records from the result set.

It also can be used with multiple columns.

`DISTINCT` applies to all selected columns, meaning the entire row must be unique to be included in the results.

It can be **slow** on large datasets because it requires sorting and filtering.

It **does not modify** the actual table data; it only affects the query results.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

```
<table>
   <thead>
      <tr>
         <th>Launch_Site</th>
      </tr>
   </thead>
   <tbody>
      <tr>
         <td>CCAFS LC-40</td>
      </tr>
      <tr>
         <td>VAFB SLC-4E</td>
      </tr>
      <tr>
         <td>KSC LC-39A</td>
      </tr>
      <tr>
         <td>CCAFS SLC-40</td>
      </tr>
   </tbody>
</table>
```

Launch Site Names Begin with 'CCA'

SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE '%CCA%' LIMIT 5

(retrieve 5 entries where the launch site name begins 'CCA')

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

The `LIKE` operator is used in SQL to search for a specific pattern within a column, usually with **text values**.

- **% (percent sign):** Matches **zero, one, or multiple** characters.
- **_ (underscore):** Matches **exactly one** character.

The `LIMIT` clause restricts the number of rows returned in a query result.

Total Payload Mass

SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE

SUM(PAYLOAD_MASS__KG_)
619967

(calculate and present the cumulative payload mass of NASA launches)

The `SUM` aggregate function in SQL is used to calculate the **total sum** of a numeric column.

✅ `SUM` works only on numeric columns.

✅ It ignores `NULL` values in the column.

✅ Often used with `GROUP BY` to summarize data

The other aggregate function are `AVG`, `MAX`, `MIN`, and `COUNT`.

Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_)
    FROM SPACEXTABLE
    WHERE Booster_Version LIKE '%F9 v1.1%'
```

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

(show the mean payload mass for launches using the F9 v1.1 booster)

Query Breakdown:

- `SELECT AVG(PAYLOAD_MASS__KG_)` → Computes the **average** of the `PAYLOAD_MASS__KG_` column.
- `FROM SPACEXTABLE` → Retrieves data from the `SPACEXTABLE`.
- `WHERE Booster_Version LIKE '%F9 v1.1%'` → Filters only rows where the `Booster_Version` **contains** `'F9 v1.1'` (due to %, which acts as a wildcard for any characters before or after).

First Successful Ground Landing Date

```sql
SELECT MIN(Date)
    FROM SPACEXTABLE
    WHERE  Landing_Outcome = 'Success (ground pad)'
```

MIN(Date)
2015-12-22

This SQL query finds the **earliest (minimum) date** when a SpaceX rocket successfully landed on a **ground pad**.

Query Breakdown:

- `SELECT MIN(Date)` → Retrieves the **earliest date** from the `Date` column.
- `FROM SPACEXTABLE` → Searches within the `SPACEXTABLE`.
- `WHERE Landing_Outcome = 'Success (ground pad)'` → Filters only rows where the **landing was successful on a ground pad**.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT Booster_Version
      FROM SPACEXTABLE
      WHERE Landing_Outcome = 'Success (drone ship)'
            AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

This SQL query retrieves the **booster versions** that successfully landed on a **drone ship** while carrying a payload between **4000 and 6000 kg**.

Query Breakdown:

- `SELECT Booster_Version` → Retrieves the **booster version** used in the launch.
- `FROM SPACEXTABLE` → Searches within the SPACEXTABLE.
- `WHERE Landing_Outcome = 'Success (drone ship)'` → Filters launches that **successfully landed on a drone ship**.
- `AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000` → Further filters results to **only include launches with a payload mass between 4000 and 6000 kg**.

Total Number of Successful and Failure Mission Outcomes

SELECT Mission_Outcome, COUNT(*) AS Total_Count
FROM SPACEXTABLE
GROUP BY Mission_Outcome

(the count of successful versus failed mission outcomes)

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Query Breakdown:

`SELECT Mission_Outcome, COUNT(*) AS Total_Count` → **Selects two fields to return:**

- `Mission_Outcome` – retrieves each unique value from the `Mission_Outcome` column.
- `COUNT(*) AS Total_Count` – counts the total number of rows (including NULLs) in each group of `Mission_Outcome`. The result is labeled as `Total_Count`.

`FROM SPACEXTABLE` → **Specifies the table** `SPACEXTABLE` from which the data will be queried.

`GROUP BY Mission_Outcome` → **Groups the result set** by each unique value in the `Mission_Outcome` column. The `COUNT(*)` function is applied to each group.

Boosters Carried Maximum Payload

```
SELECT Booster_Version, PAYLOAD_MASS__KG_
    FROM SPACEXTABLE
    WHERE PAYLOAD_MASS__KG_= (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE)
```

This query retrieves the Booster Version and Payload Mass (in kilograms) from the SPACEXTABLE, but only for the row where the Payload Mass is the maximum value in the table.

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Breakdown of the query:

1. **Subquery** (`select MAX(PAYLOAD_MASS__KG_) from SPACEXTABLE`) → finds the maximum value of the **PAYLOAD_MASS__KG_** column in the **SPACEXTABLE**.
2. **Main Query** (`select Booster_Version, PAYLOAD_MASS__KG_ from SPACEXTABLE where PAYLOAD_MASS__KG_ = ...`) → selects the **Booster Version** and **Payload Mass (kg)** from the table where the **Payload Mass** equals the maximum value found in the subquery.

2015 Launch Records

```
SELECT substr(Date, 6,2) as Month, Landing_Outcome, Booster_Version, Launch_Site
    from SPACEXTABLE
        where Landing_Outcome='Failure (drone ship)'
            and substr(Date,0,5)='2015'
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

This query retrieves the **Month**, **Landing Outcome**, **Booster Version**, and **Launch Site** from the **SPACEXTABLE** for launches that met two specific conditions:

1. **Landing Outcome = 'Failure (drone ship)'**: The landing failed on a drone ship.
2. **Year = 2015**: The launch occurred in the year 2015.

Breakdown of the query:

- `substr(Date, 6,2) as Month`: Extracts the month (the 6th and 7th characters) from the **Date** column and labels it as **Month**
- `substr(Date, 0, 5)`: Extracts the first 4 characters from the **Date** column, which represents the year, and checks if it's **'2015'**
- `Landing_Outcome = 'Failure (drone ship)'`: Filters the rows where the landing outcome was a failure on a drone ship

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
select Landing_Outcome, Date, COUNT(*) as Landing_Outcome_count
    from SPACEXTABLE
        where Date between '2010-06-04' and '2017-03-20'
        GROUP BY Landing_Outcome
        ORDER BY Landing_Outcome_count DESC
```

Landing_Outcome	Date	Landing_Outcome_count
No attempt	2012-05-22	10
Success (drone ship)	2016-04-08	5
Failure (drone ship)	2015-01-10	5
Success (ground pad)	2015-12-22	3
Controlled (ocean)	2014-04-18	3
Uncontrolled (ocean)	2013-09-29	2
Failure (parachute)	2010-06-04	2
Precluded (drone ship)	2015-06-28	1

This query retrieves the **Landing Outcome**, the **Date** of the landing, and the **count** of each landing outcome from the **SPACEXTABLE**, for launches that occurred between **June 4, 2010** and **March 20, 2017**.

Breakdown of the query:

1. `WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'`: Filters the rows to include only those with a **Date** between June 4, 2010, and March 20, 2017
2. `GROUP BY Landing_Outcome`: Groups the results by **Landing Outcome**, so the query will count the number of occurrences for each landing outcome
3. `COUNT(*) as Landing_Outcome_count`: Counts how many times each **Landing Outcome** appears in the selected date range
4. `ORDER BY Landing_Outcome_count DESC`: Orders the results by the count of each landing outcome in descending order, showing the most frequent outcomes first

Section 3

Launch Sites Proximities Analysis

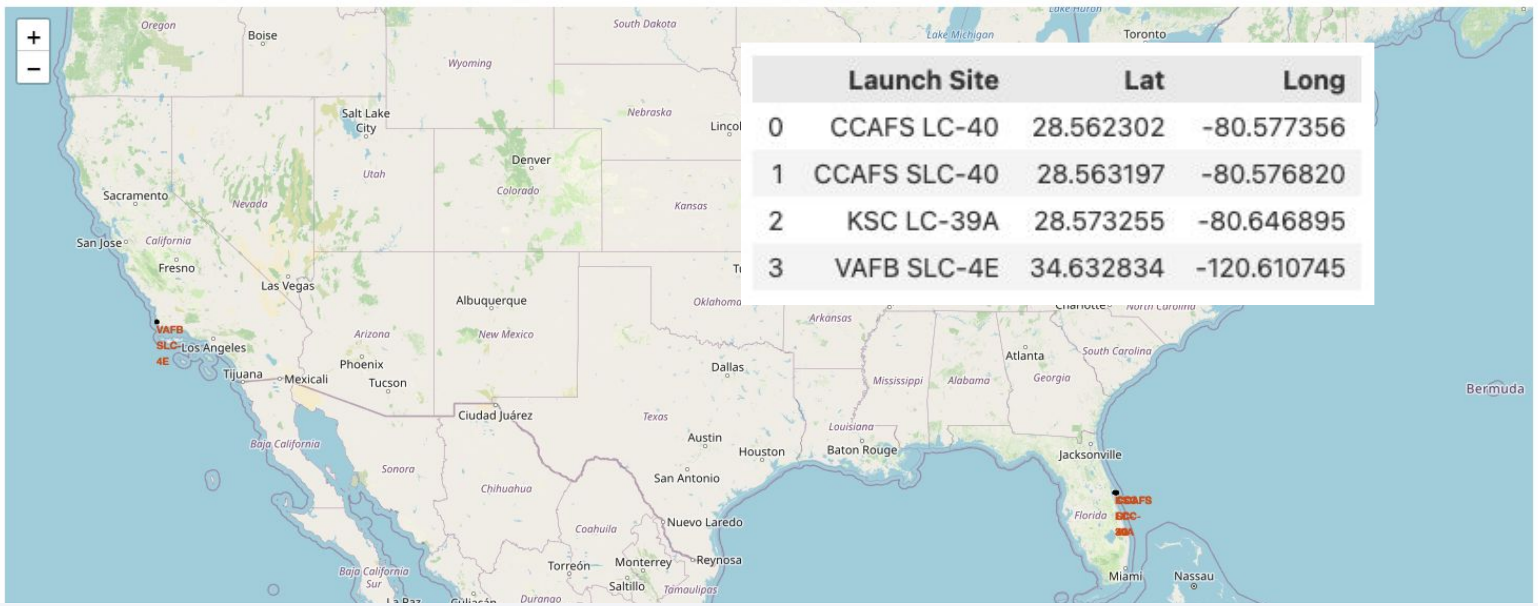
All launch sites on a map

CCAFS LC-40, CCAFS SLC-40 and KSC LC-39A are situated in Florida.

VAFB SLC-4E is situated in California.

While being closer to the Equator is beneficial for launching satellites into geostationary orbit (due to Earth's rotational speed), the U.S. launch sites are still at moderate latitudes.

The **Cape Canaveral** sites are closer to the Equator (~28° N) than **Vandenberg** (~34° N), which is much farther north and is mainly used for polar orbits rather than equatorial ones.



	Launch Site	Lat	Long
0	CCAFS LC-40	28.562302	-80.577356
1	CCAFS SLC-40	28.563197	-80.576820
2	KSC LC-39A	28.573255	-80.646895
3	VAFB SLC-4E	34.632834	-120.610745

All launch sites are located near the coast. The given coordinates show that all sites are **within a few kilometers of the ocean**.

- Coastal locations allow rockets to launch over the ocean, **reducing the risk of debris falling on populated areas**.
- **Cape Canaveral (Florida)** launches mostly go eastward over the Atlantic Ocean.
- **Vandenberg (California)** launches mainly go southward over the Pacific Ocean, ideal for polar orbits.

Conclusion

✅ **Launch sites are not close to the Equator but are in mid-latitudes.**
✅ **All launch sites are near the coast for safety reasons and optimal launch trajectories.**

Success / failed launches for each site on the map

From the color-labeled markers in marker clusters, it’s identified that launch site KSC LC-39A has relatively high success rates.

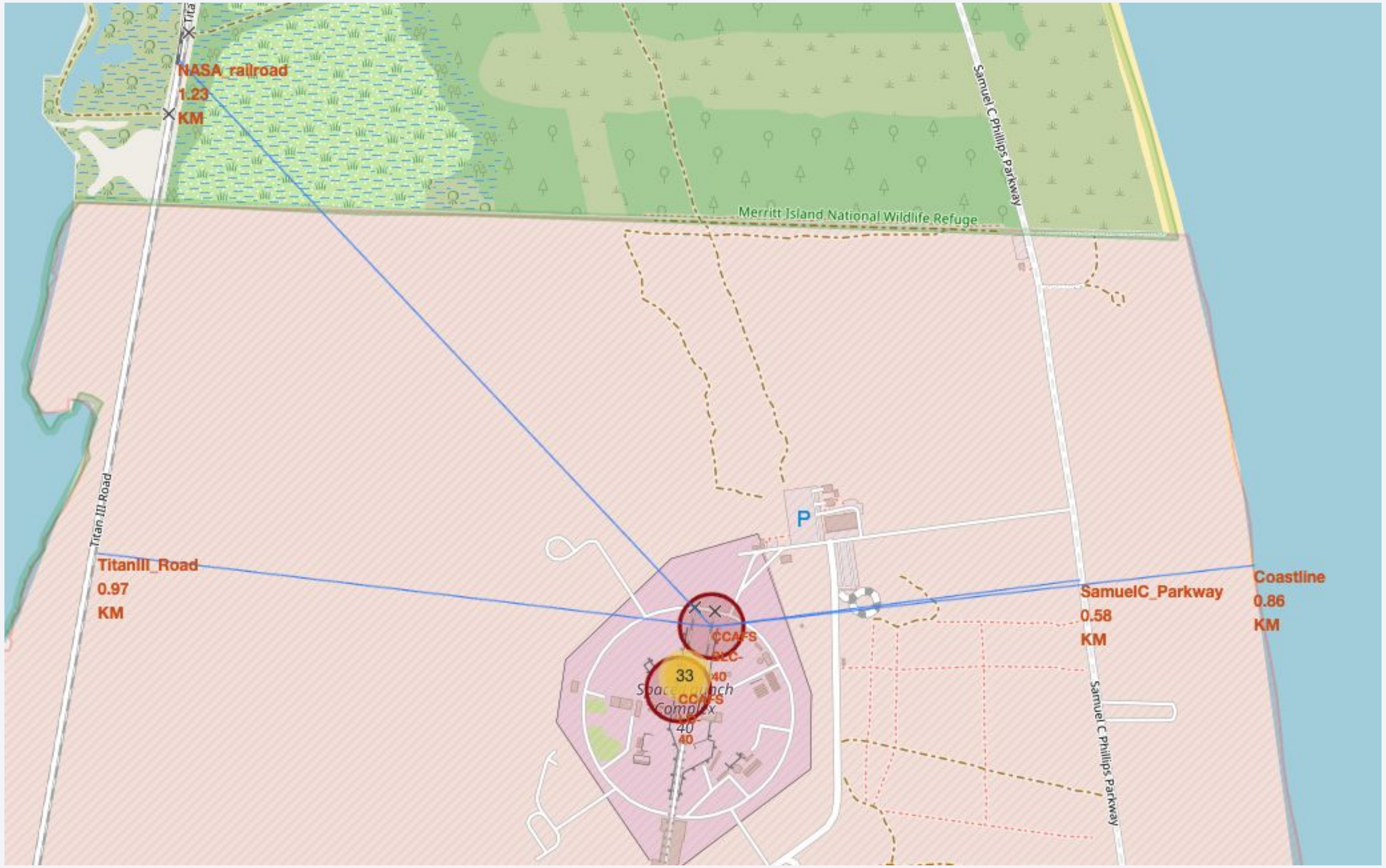
Distance between a launch site to its proximities

- NASA Railroad: 1.23 KM (northwest)
- Titan III Road: 0.97 KM (southwest)
- Samuel C. Parkway: 0.58 KM (east)
- Coastline: 0.86 KM (east)

✅ **Launch sites are located near the coast** for safety reasons, allowing failed rocket stages to fall into the ocean instead of populated areas.

✅ **Infrastructure like roads and railroads are present** to transport rocket components and fuel.

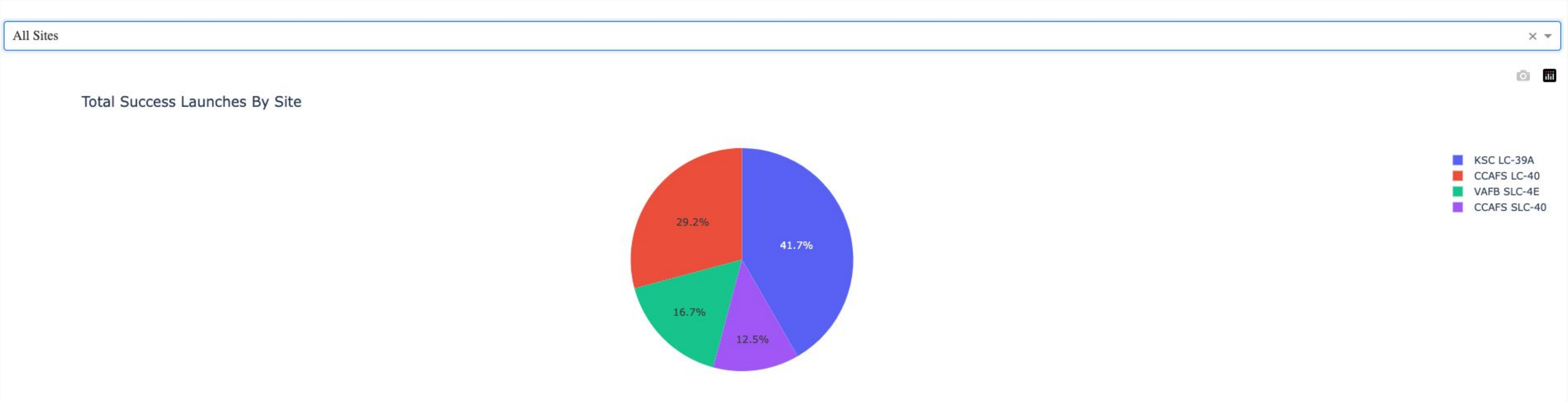
✅ **The restricted zone (shaded area)** ensures no unauthorized personnel are close to the launch pad during operations.



Section 4

Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard



✅ **KSC LC-39A has the highest success rate (41.7%)**, making it the most reliable launch site.

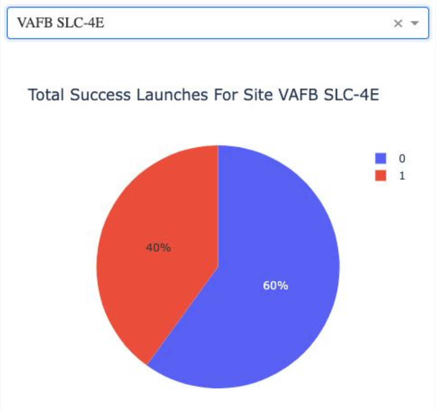
✅ **CCAFS LC-40 follows with 29.2% of successful launches.**

✅ **VAFB SLC-4E and CCAFS SLC-40 have lower success percentages (16.7% and 12.5%).**

✅ **The dashboard allows filtering by launch site, helping analyze performance trends individually.**

Launch site with highest launch success ratio

Blue (1) → Successful launches
Red (0) → Unsuccessful launches

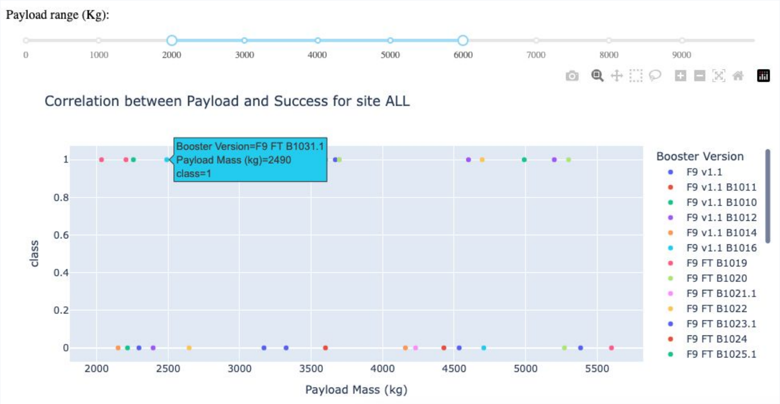


✅ **KSC LC-39A has a high success rate (76.9%)**, making it a reliable launch site.

✅ **Failure rate is 23.1%**, indicating some unsuccessful launches but overall strong performance.

✅ **The filtering function allows analyzing specific sites separately, which is useful for trend evaluation.**

Payload vs. Launch Outcome scatter plot for all sites



✅ Payload range 3000-4000 kg has the highest launch success rate.

✅ Payload range 6000-7000 kg has the lowest launch success rate

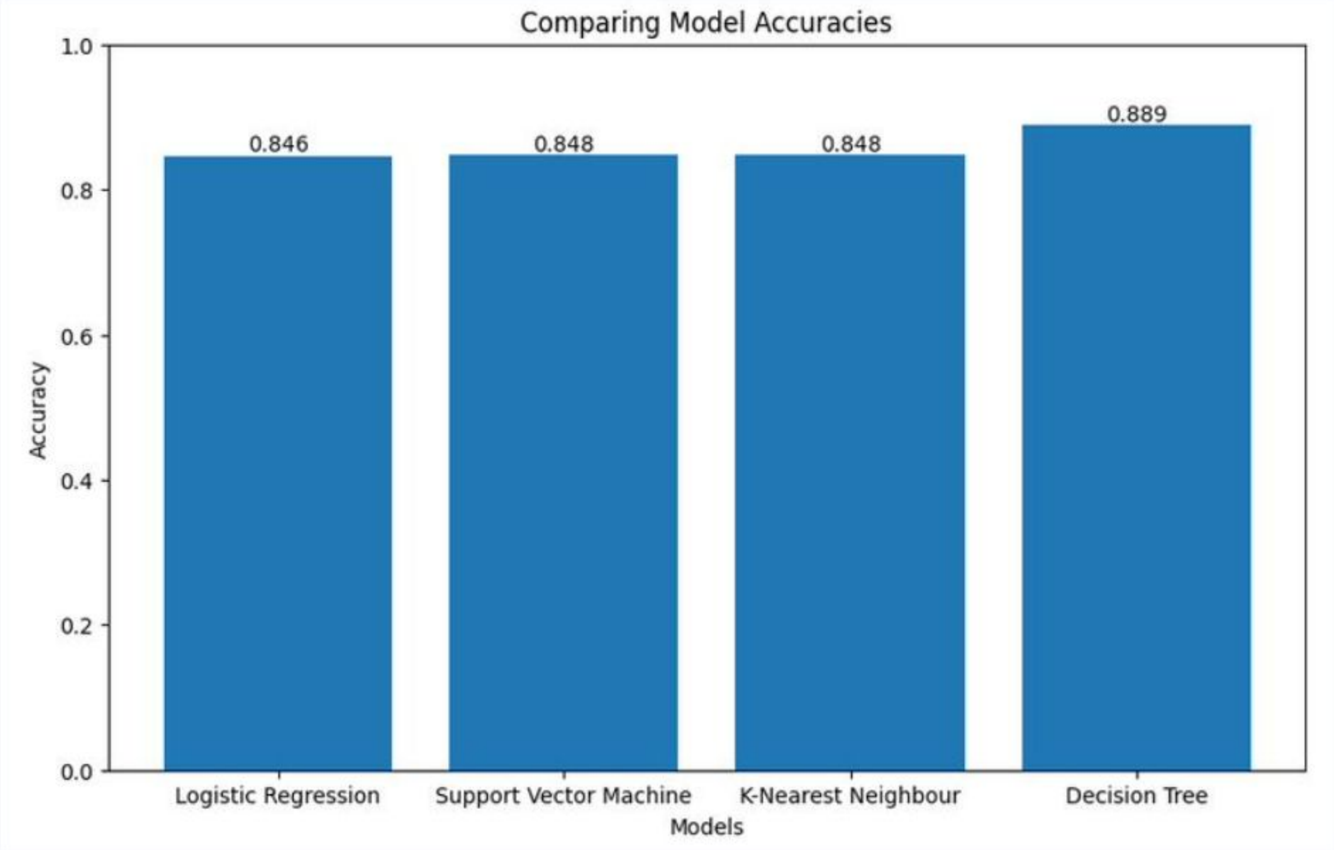
✅ Booster versions FT and B5 have the highest launch success rate 65% and 100%, respectively. But B5 has only 1 launch. At the same time FT has 23 launches (15:8)

Section 5

Predictive Analysis (Classification)

Classification Accuracy

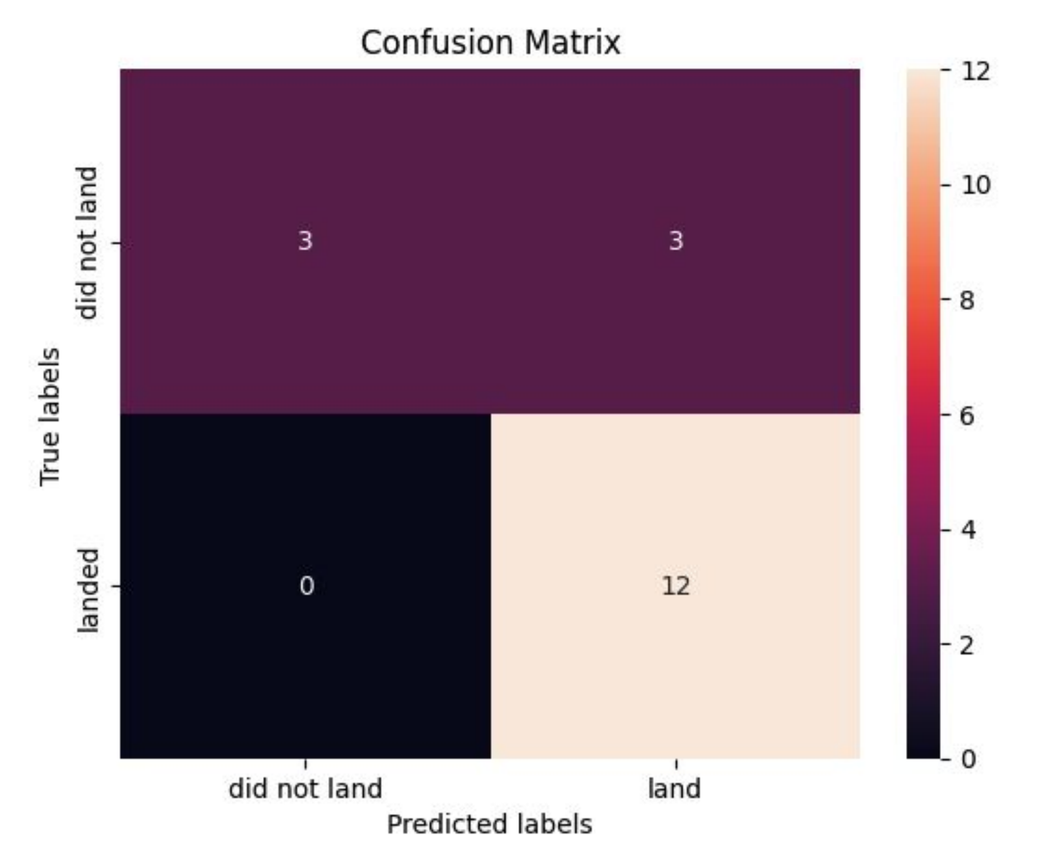
The Decision Tree model achieved the highest accuracy of 0.889, while Logistic Regression, Support Vector Machine, and K-Nearest Neighbour models have slightly lower accuracies, each at around 0.84



Confusion Matrix

All 4 model demonstrate the same confusion matrix. Each of them can distinguish between the different classes.

	Predicted: Did Not Land	Predicted: Land
Actual: Did Not Land	3 (True Negative)	3 (False Positive)
Actual: Landed	0 (False Negative)	12 (True Positive)



Key Observations:

- **True Positives (TP) = 12** → The model correctly predicted 12 landings.
- **True Negatives (TN) = 3** → The model correctly identified 3 non-landings.
- **False Positives (FP) = 3** → The model mistakenly predicted landing when it didn't happen.
- **False Negatives (FN) = 0** → The model never misclassified a landed rocket as not landed.

Conclusions

- The Decision Tree model demonstrates strong predictive capabilities for successful landings but exhibits limitations in accurately classifying failed landing attempts.
- Notably, the frequency of successful landings has increased over time, likely due to ongoing technical enhancements
- KSC LC-39A stands out as the most successful among SpaceX's launch sites
- Orbits ES-L1, GEO, HEO, and SSO demonstrate a 1.0 success rate, indicating perfect landing
  success for all launches within these orbits
- The majority of launch sites are strategically situated in coastal proximity and near the equator to optimize launch efficiency and safety

Thank you!